

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: EcoCart – Sustainable Products E-Commerce Order Management System

Skill Task: DBMS

Week: 4

1. Database Design

1.1 Orders Table

Attribute	Data Type	Constraint
<i>Order_ID</i>	INT	Primary Key
<i>Customer_ID</i>	INT	Foreign Key
<i>Order_Date</i>	DATE	NOT NULL
<i>Total_Amount</i>	NUMBER(10,2)	NOT NULL

1.2 Order_Details Table

Attribute	Data Type	Constraint
<i>Order_Detail_ID</i>	INT	Primary Key
<i>Order_ID</i>	INT	Foreign Key
<i>Product_ID</i>	INT	Foreign Key
<i>Quantity</i>	INT	NOT NULL
<i>Unit_Price</i>	NUMBER(10,2)	NOT NULL
<i>Subtotal</i>	NUMBER(10,2)	NOT NULL

2. Table Relationships



3. Table Creation

3.1 Create Orders Table

```
SQL> CREATE TABLE Orders (  
2     Order_ID      NUMBER(5) PRIMARY KEY,  
3     Customer_ID   NUMBER(5),  
4     Order_Date    DATE NOT NULL,  
5     Total_Amount  NUMBER(10,2) NOT NULL,  
6     CONSTRAINT fk_orders_customer  
7     FOREIGN KEY (Customer_ID)  
8     REFERENCES Customer(Customer_ID)  
9 );
```

Table created.

3.2 Create Order_Details

```
SQL> CREATE TABLE Order_Details (  
2     Order_Detail_ID NUMBER(5) PRIMARY KEY,  
3     Order_ID        NUMBER(5),  
4     Product_ID      NUMBER(5),  
5     Quantity        NUMBER(5) NOT NULL,  
6     Unit_Price      NUMBER(10,2) NOT NULL,  
7     Subtotal        NUMBER(10,2) NOT NULL,  
8     CONSTRAINT fk_order_details_order  
9     FOREIGN KEY (Order_ID)  
10    REFERENCES Orders(Order_ID),  
11    CONSTRAINT fk_order_details_product  
12    FOREIGN KEY (Product_ID)  
13    REFERENCES Product(Product_ID)  
14 );
```

Table created.

4. Order Data Insertion

```
SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES
  4 (201, 101, SYSDATE, 2500.00);

1 row created.

SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES
  4 (202, 102, TO_DATE('11-SEP-2026', 'DD-MON-YYYY'), 1800.00);

1 row created.

SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES
  4 (203, 103, TO_DATE('12-SEP-2026', 'DD-MON-YYYY'), 3200.00);

1 row created.

SQL> INSERT INTO Orders
  2 (Order_ID, Customer_ID, Order_Date, Total_Amount)
  3 VALUES
  4 (204, 104, TO_DATE('13-SEP-2026', 'DD-MON-YYYY'), 1500.00);

1 row created.
```

5. Order_Details Data Insertion

```
SQL> INSERT INTO Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (3, 202, 103, 2, 900.00, 1800.00);

1 row created.

SQL> INSERT INTO Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (4, 203, 104, 2, 1600.00, 3200.00);

1 row created.

SQL> INSERT INTO Order_Details
  2 (Order_Detail_ID, Order_ID, Product_ID, Quantity, Unit_Price, Subtotal)
  3 VALUES
  4 (5, 204, 101, 1, 1500.00, 1500.00);

1 row created.
```

6. Order Modification Operations

6.1 Modify Order Amount

```
SQL> UPDATE Orders
  2  SET Total_Amount = 2700.00
  3  WHERE Order_ID = 201;

1 row updated.
```

```
SQL> SELECT * FROM Orders
  2  WHERE Order_ID = 201;

ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
-----
      201          101 16-SEP-26          2700
```

6.2 Modify Order Date

```
SQL> UPDATE Orders
  2  SET Order_Date = TO_DATE('15-SEP-2026', 'DD-MON-YYYY')
  3  WHERE Order_ID = 202;

1 row updated.
```

7. Order_Details Modification

```
SQL> UPDATE Order_Details
  2  SET Quantity = 3,
  3      Subtotal = 3000.00
  4  WHERE Order_Detail_ID = 1;

1 row updated.
```

8. Customer Order History Report

```
SQL> SELECT
  2     c.Customer_ID,
  3     c.Customer_Name,
  4     o.Order_ID,
  5     o.Order_Date,
  6     od.Product_ID,
  7     od.Quantity,
  8     od.Unit_Price,
  9     od.Subtotal
 10 FROM Customer c
 11 JOIN Orders o
 12     ON c.Customer_ID = o.Customer_ID
 13 JOIN Order_Details od
 14     ON o.Order_ID = od.Order_ID
 15 ORDER BY c.Customer_ID, o.Order_Date;
```

CUSTOMER_ID

CUSTOMER_NAME

ORDER_ID	ORDER_DAT	PRODUCT_ID	QUANTITY	UNIT_PRICE	SUBTOTAL
101					
Anitha Kumar	201 16-SEP-26	101	3	1000	3000
101					
Anitha Kumar	201 16-SEP-26	102	1	500	500

9. Customer-wise Order Summary

```
SQL> SELECT
  2     c.Customer_ID,
  3     c.Customer_Name,
  4     COUNT(o.Order_ID) AS Total_Orders,
  5     SUM(o.Total_Amount) AS Total_Amount
  6 FROM Customer c
  7 JOIN Orders o
  8     ON c.Customer_ID = o.Customer_ID
  9 GROUP BY c.Customer_ID, c.Customer_Name
 10 ORDER BY c.Customer_ID;
```

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

101		
Anitha Kumar	1	2700
102		
Rahul	1	1800

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

101		
Anitha Kumar	1	2700
102		
Rahul	1	1800

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

103		
Priya	1	3200

104

Karthik S

CUSTOMER_ID

CUSTOMER_NAME

TOTAL_ORDERS TOTAL_AMOUNT

1	1500
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